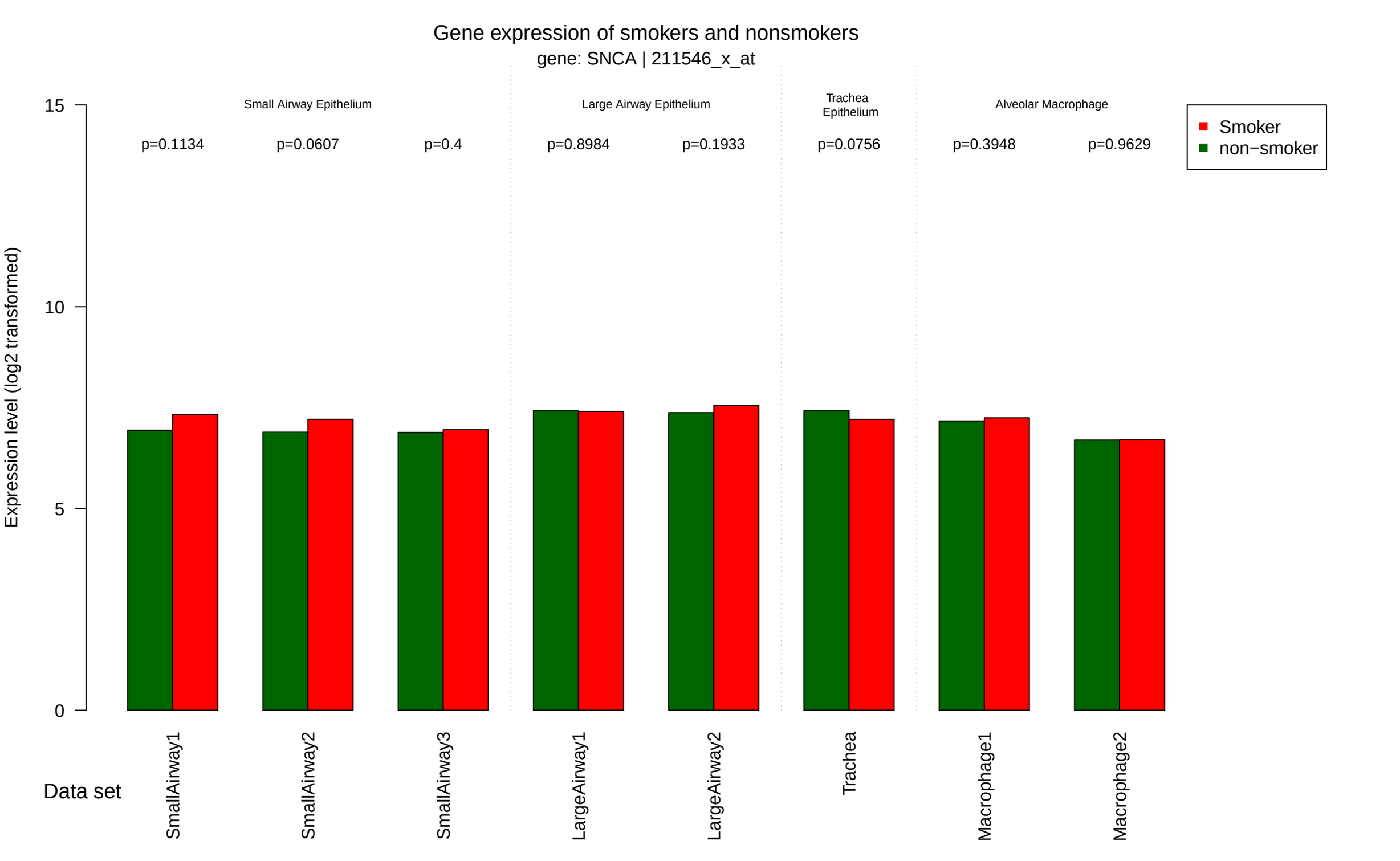
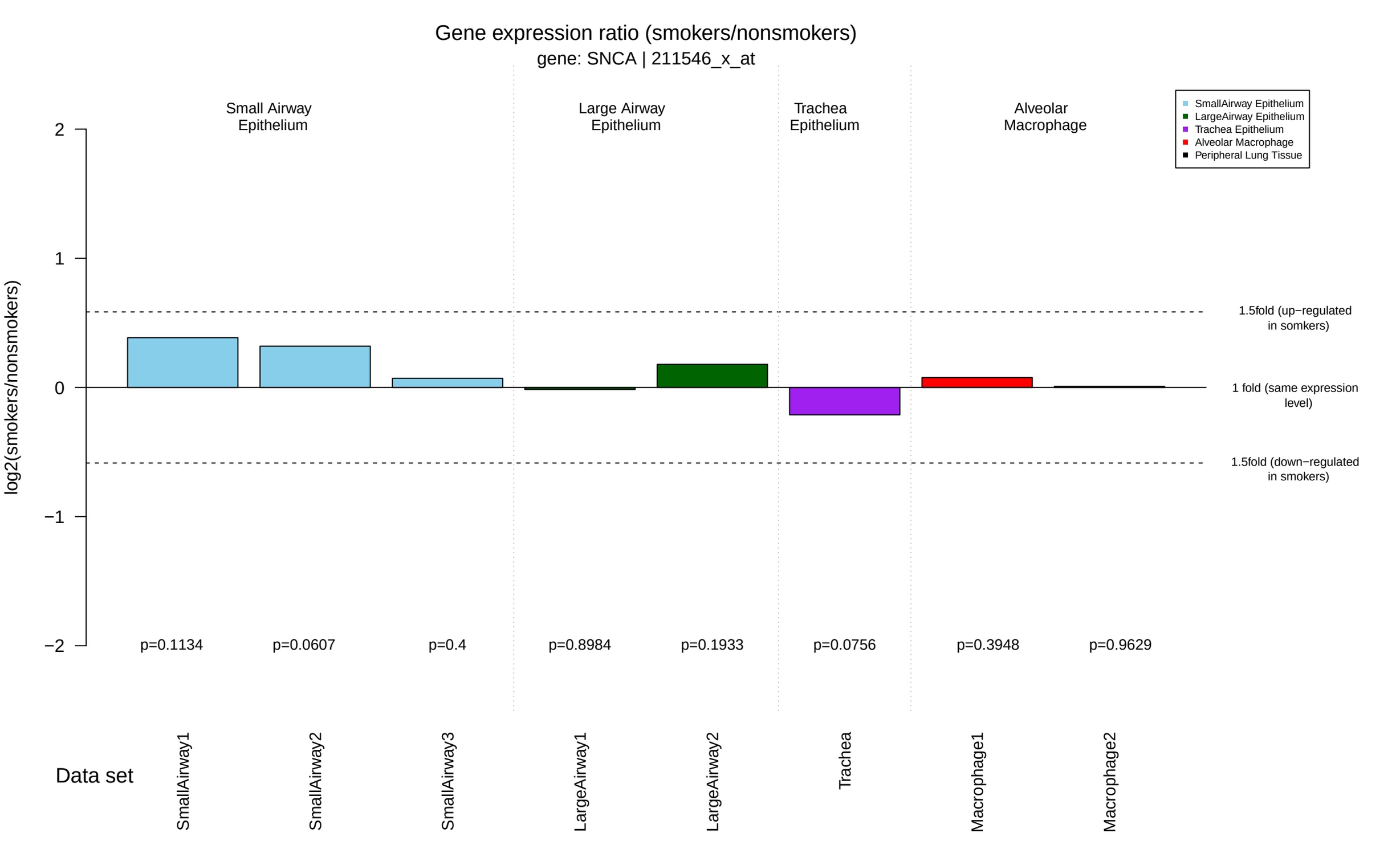
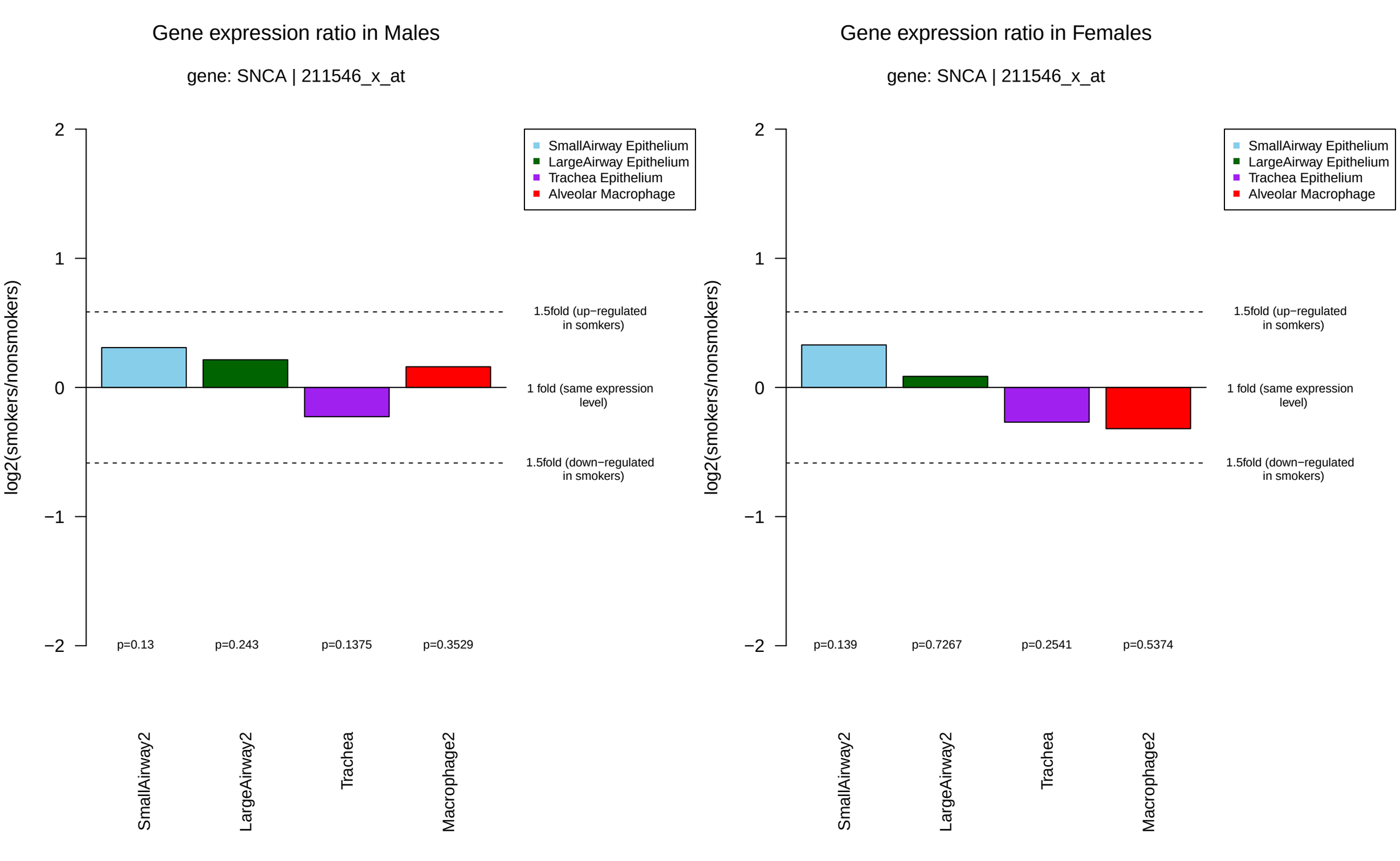
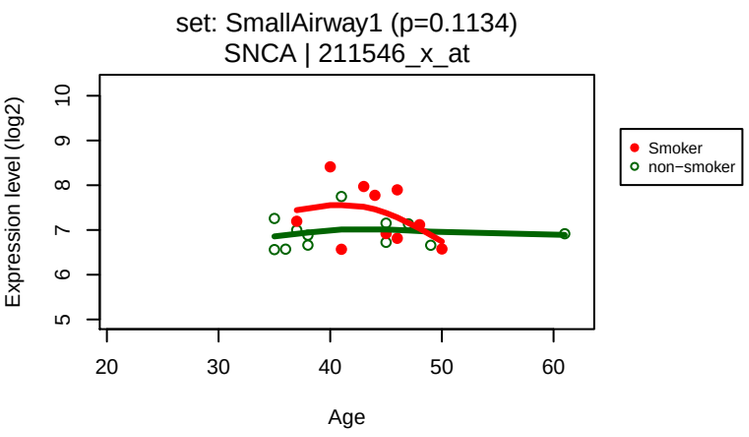
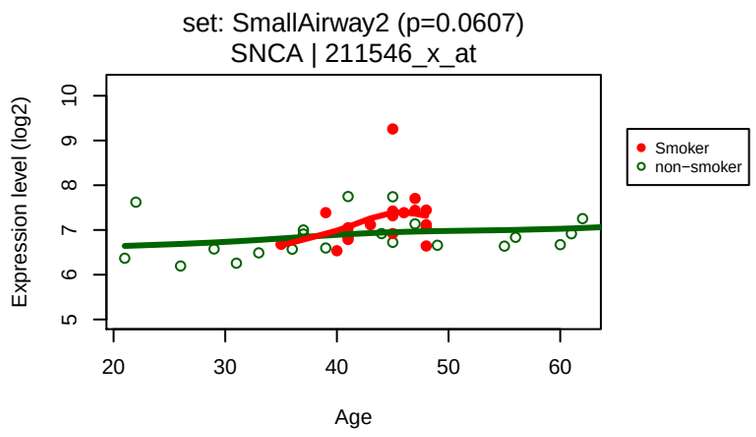
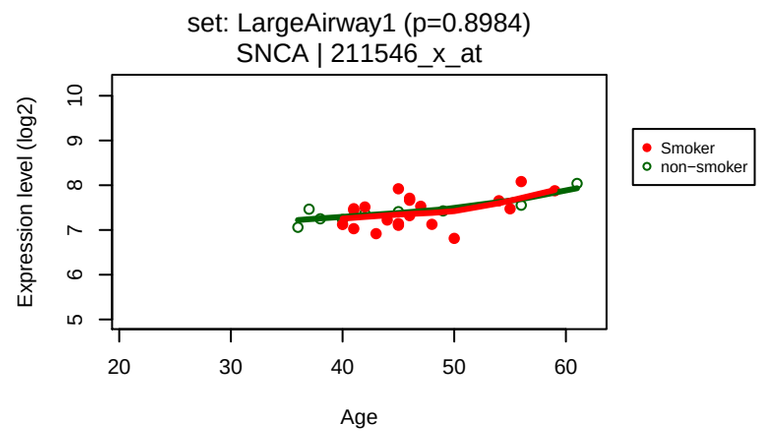

Gene expression of smokers and nonsmokers
gene: SNCA | 211546_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1134
p=0.0607
p=0.4
p=0.8984
p=0.1933
p=0.0756
p=0.3948
p=0.9629
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

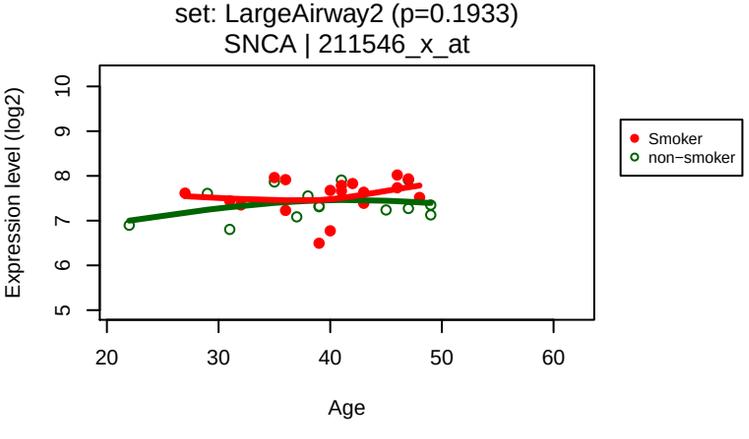

Gene expression ratio (smokers/nonsmokers)
gene: SNCA | 211546_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1134
p=0.0607
p=0.4
p=0.8984
p=0.1933
p=0.0756
p=0.3948
p=0.9629
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

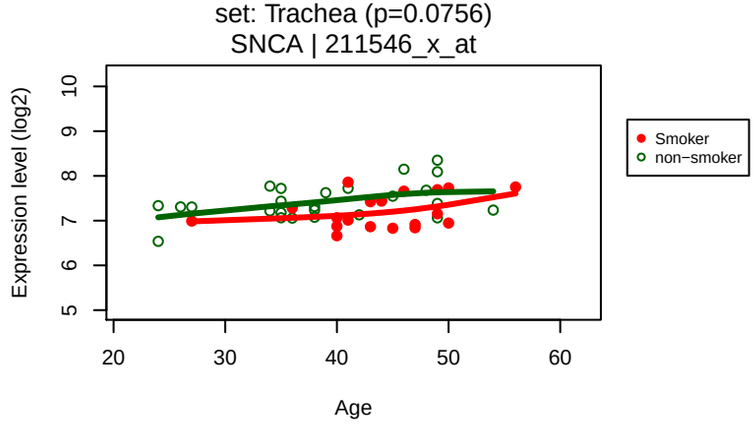

Gene expression ratio in Males
gene: SNCA | 211546_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.13
p=0.243
p=0.1375
p=0.3529
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SNCA | 211546_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.139
p=0.7267
p=0.2541
p=0.5374
SmallAirway2
LargeAirway2
Trachea
Macrophage2

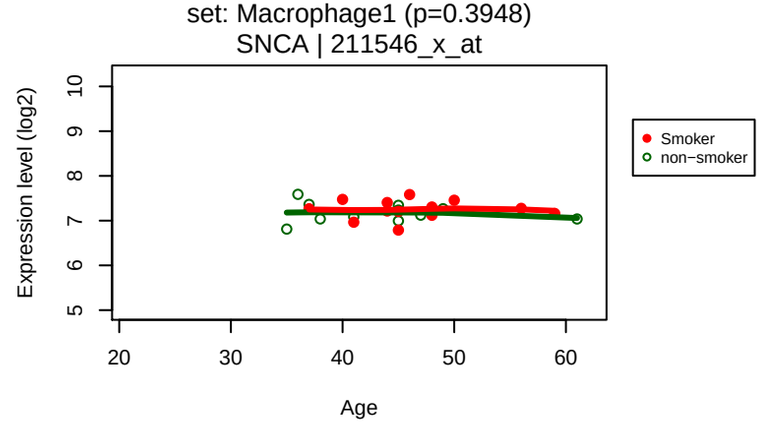

set: SmallAirway1 (p=0.1134)
SNCA | 211546_x_at
Expression level (log2)
Age
Smoker
non−smoker

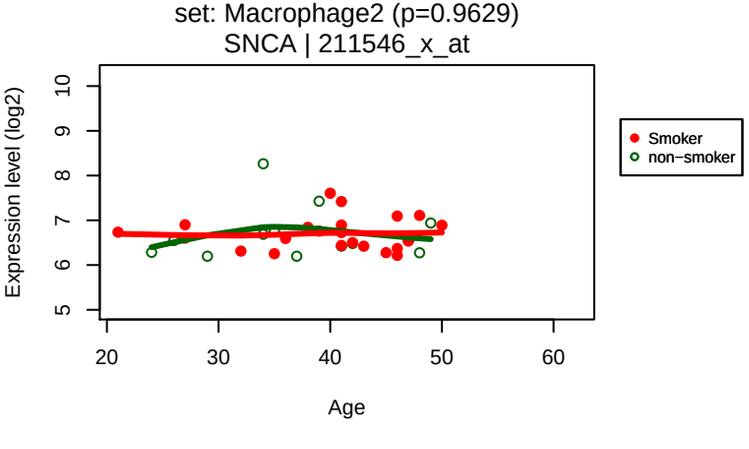

set: SmallAirway2 (p=0.0607)
SNCA | 211546_x_at
Expression level (log2)
Age
Smoker
non−smoker

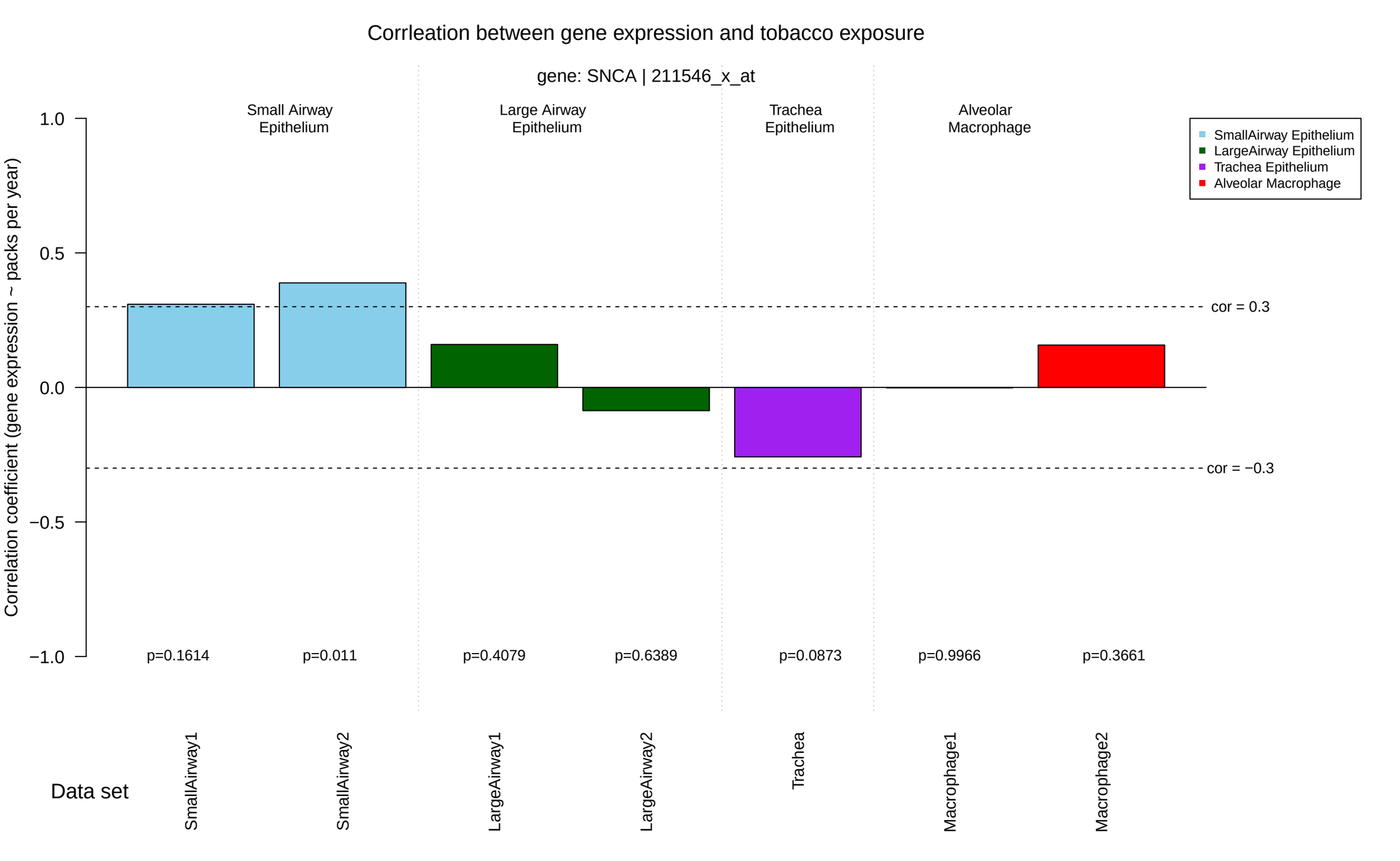

set: LargeAirway1 (p=0.8984)
SNCA | 211546_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1933)
SNCA | 211546_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0756)
SNCA | 211546_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3948)
SNCA | 211546_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9629)
SNCA | 211546_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SNCA | 211546_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1614
p=0.011
p=0.4079
p=0.6389
p=0.0873
p=0.9966
p=0.3661
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2